

Yariv Barsheshat

SOFTWARE ENGINEERING · AI & MACHINE LEARNING · SOLUTIONS ARCHITECTURE

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Experience

Independent AI Safety Researcher

Montréal, QC

FUNDED BY COEFFICIENT GIVING

Apr 2026 – Present

- Conducting independent AI safety research in collaboration with Robert Graham, supported by a research grant from Coefficient Giving.
- Co-authored a research paper accepted to two ICML 2026 workshops (AI4GOOD and Pluralistic Alignment) and currently under review at NeurIPS 2026; topic and abstract under embargo pending presentation.

AI Full-Stack Engineer

Montréal, QC

EXPLORAI

Jan 2026 – Apr 2026

- Contributed across the full stack to Estimai (www.estim.ai), a SaaS platform enabling construction estimators to generate bids on calls for tender by ingesting architectural plans and technical specs via AI-driven document analysis.
- Designed and single-handedly delivered a PoC for detecting and associating column icons across multi-page floor plans using vectorial detection algorithms on extracted PDF primitives.
- Helped shape technology and design decisions, balancing scalability, cost-efficiency, and rapid iteration for a cloud-deployed product.

Senior Research & Development Programmer

Montréal, QC

MTL.AI

Aug 2022 – Jan 2026

- Designed, trained, and deployed real-time computer-vision models for object detection and segmentation in live soccer broadcasts, powering ad replacement for UEFA Euro 2024 and Champions League.
- Identified and resolved key pipeline bottlenecks, achieving a 5× speedup in model validation and accelerating deployment cycles.
- Built a regional threshold-control feature giving operators fine-grained control over model behaviour, reducing recurring spatial errors during live broadcasts.
- Led development of a unified API abstraction library integrating frontier LLMs (OpenAI, Google, Anthropic) into a production marketplace assistant app, enabling rapid model swapping and cost optimization. (see www.tavendi.com)

Mathematics Professor

Montréal, QC

DAWSON COLLEGE

Jun 2015 – Jul 2022

- Taught calculus, linear algebra, statistics, and probability to thousands of students; developed course materials adopted department-wide. Executive council member on the Teachers' Union and Science Program Committee.

Education

McGill University

Montréal, QC

MASTER OF SCIENCE IN MATHEMATICS

2013 – 2015

- Thesis: *Entropy in Dynamical Systems* — shortlisted for top Master's thesis in Mathematics.

McGill University

Montréal, QC

BACHELOR OF SCIENCE, JOINT HONOURS IN MATHEMATICS AND PHYSICS

2009 – 2012

- Dean's Honour List — GPA 3.91

Skills

Languages Python, C++, Kotlin, Java, SQL, HTML, $\text{\LaTeX}$

Frameworks & Tools PyTorch, OpenCV, CUDA, TensorRT, FastAPI, Pydantic, Docker, Git, Jira, Confluence

Architecture REST API design, microservice architecture, CI/CD pipelines, cloud deployment

Spoken Languages English (Native), French (Fluent), Hebrew (Fluent)

Selected Awards

2014, 2015 **NSERC Canada Graduate Scholarship**, Government of Canada

McGill University

2010, 2011 **NSERC Undergraduate Student Research Award**, Government of Canada

McGill University